

11 gravitational force provides centripetal force

$$G \frac{Mm}{r^2} = \frac{mv^2}{r}$$

$$\begin{aligned} M &= \frac{rv^2}{G} = \frac{(5.7 \times 10^{17})(7.5 \times 10^5)^2}{6.67 \times 10^{-11}} \\ &= 4.8 \times 10^{39} \text{ kg} \end{aligned}$$